



Design and Development of Darunavir Loaded Self Micro Emulsifying Drug Delivery System using Extreme Vertices Mixture Design in a Quality by Design Framework

Purpose: The purpose of this research paper is to develop and optimize a Darunavir-loaded self-micro emulsifying drug delivery system (SMEDDS) using an Extreme Vertices Mixture Design within a Quality by Design (QbD) framework. The mixture components examined are Capryol 90, Tween 80, and Transcutol HP, while the responses are drug release, drug loading, emulsification time, and droplet size.

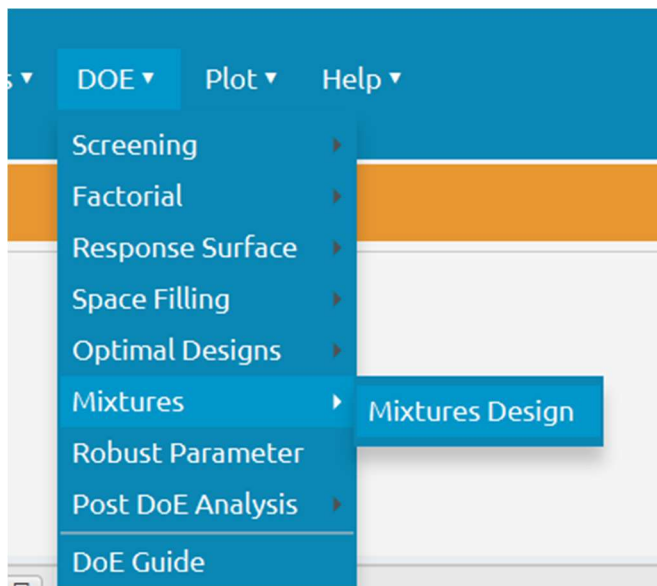
Isalos version used: 2.2.0

Scientific article:

Reddy, K. A. N., Kunchithapatham, J., Dang, R., & Ramnarayanan, C. (2020). *Design and Development of Darunavir Loaded Self Micro Emulsifying Drug Delivery System using Extreme Vertices Mixture Design in a Quality by Design Framework.*

Step 1: Extreme Vertices Design

Rename the first tab from "Action" to "Extreme Vertices". Then, go to DOE → Mixtures → Mixtures Design and select Extreme Vertices. Set the number of components to 3 and the degree of design to 2. Define the component bounds as A = 0.12–0.31, B = 0.51–0.67, and C = 0.18–0.23. Set the number of replicates and blocks to 1. Keep Random Standard Order disabled and do not select Axial Points or Center Points. Click Execute to generate the experimental design.



DoE Mixtures

Number of Replicates: 1

Number of Blocks: 1

☐ Random Standard order

Select Design: Extreme Vertices

Linear Constraints: ☐

Number of components in the Mixture: 3

Degree of Design: 2

Augment the Design with Axial Points: ☐

Augment the Design with Center Points: ☐

Component # A: Min 0.12, Max 0.31, Select Values: 0.15

Component # B: Min 0.51, Max 0.67, Select Values: 0.55

Component # C: Min 0.18, Max 0.23, Select Values: 0.2

Execute Cancel

Results (right spreadsheet) :

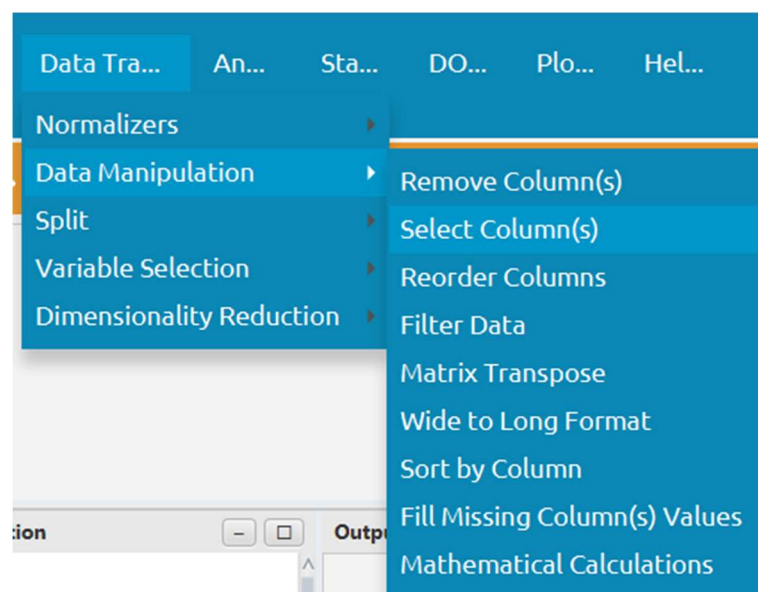
Output								
	Col1	Col2 (I)	Col3 (S)	Col4 (S)	Col5 (S)	Col6 (D)	Col7 (D)	Col8 (D)
User Header	User Row ID	Standard Order	Block Number	Replicate Number	Point Type	A	B	C
1		1	Block: 1	Replicate: 1	Design Point	0.3100000	0.51	0.18
2		2	Block: 1	Replicate: 1	Design Point	0.26	0.51	0.23
3		3	Block: 1	Replicate: 1	Design Point	0.1500000	0.67	0.18
4		4	Block: 1	Replicate: 1	Design Point	0.12	0.67	0.21
5		5	Block: 1	Replicate: 1	Design Point	0.12	0.65	0.23
6		6	Block: 1	Replicate: 1	Design Point	0.19	0.5800000	0.23
7		7	Block: 1	Replicate: 1	Design Point	0.1350000	0.67	0.195
8		8	Block: 1	Replicate: 1	Design Point	0.2850000	0.51	0.2050000
9		9	Block: 1	Replicate: 1	Design Point	0.2300000	0.5900000	0.18
10		10	Block: 1	Replicate: 1	Design Point	0.12	0.66	0.22

Step 2: Definition of Response Variables

Create a new spreadsheet and rename it “Responses”. Enter the response values reported in the paper for the 10 experimental formulations. The four response variables are Drug Release, Drug Loading, Emulsification Time, and Droplet Size.

After entering the data, go to Data Transformation → Data Manipulation → Select Column(s) and select the four response columns. Click Execute to complete the response dataset.

Input					
	Col1	Col2 (D)	Col3 (D)	Col4 (D)	Col5 (D)
User Header	User Row ID	Drug_Release	Drug_Loading	Emulsification_Time	Droplet_Size
1		79.91	97.44	45.67	260
2		77.4	97.0	55.67	278.2
3		86.14	98.0	41.33	248.53
4		87.11	98.44	42.33	253.67
5		88.8	98.0	36.67	245.1
6		74.24	97.11	57.0	290.73
7		90.64	99.0	36.0	219.9
8		85.89	98.33	45.33	257
9		93.68	100.11	26.33	188.7
10		89.92	100.11	34.33	227.2



Select Column(s)

☒ Select Columns Manually
 ☐ Select Matching Columns from Node
 ☐ Select Columns Manually and from Node

Select Node

Excluded Columns

>>

>

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Included Columns

Col2 -- Drug_Release
 Col3 -- Drug_Loading
 Col4 -- Emulsification_Time
 Col5 -- Droplet_Size

Execute

Cancel

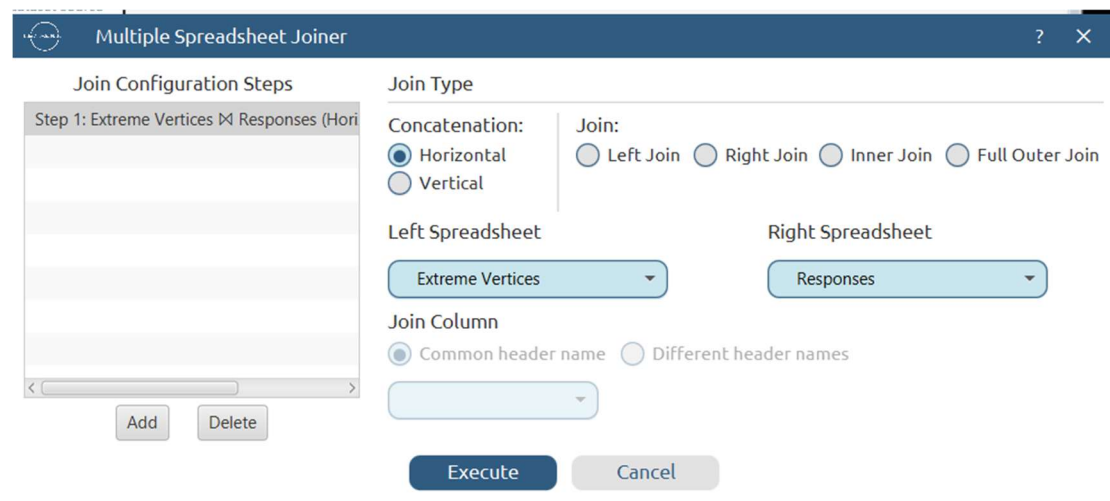
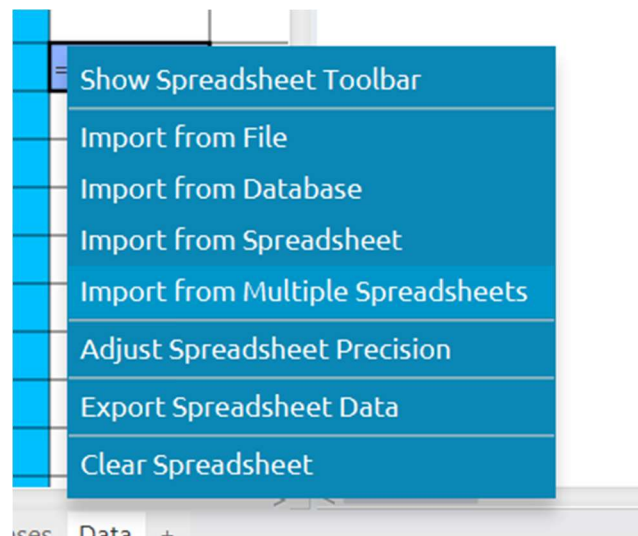
Output					
	Col1	Col2 (D)	Col3 (D)	Col4 (D)	Col5 (D)
User Header	User Row ID	Drug_Release	Drug_Loading	Emulsification_Time	Droplet_Size
1		79.91	97.44	45.67	260
2		77.4	97.0	55.67	278.2
3		86.14	98.0	41.33	248.53
4		87.11	98.44	42.33	253.67
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10		89.92	100.11	34.33	227.2
11					

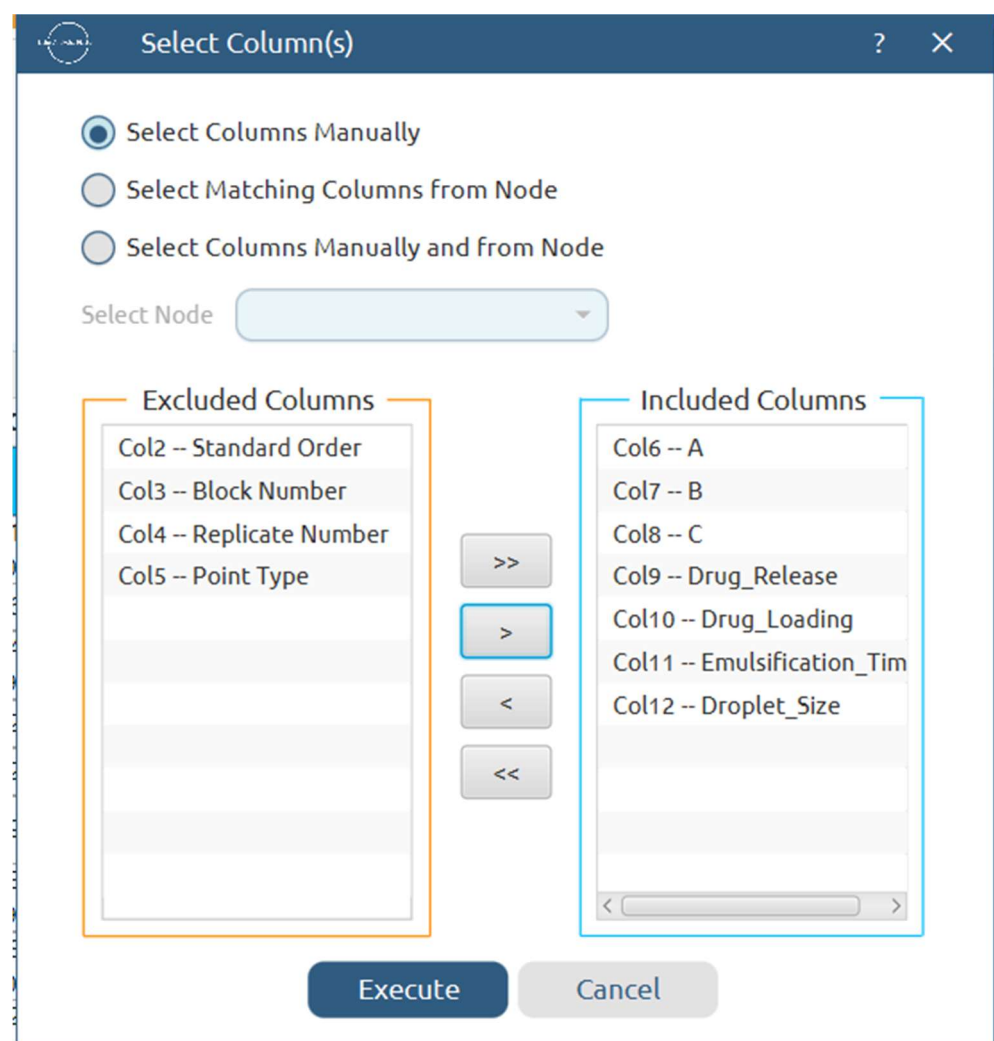
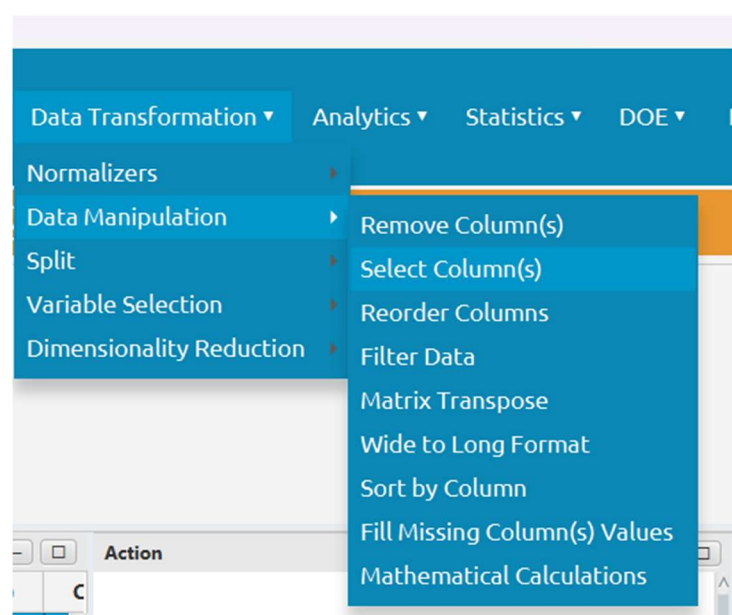
Step 3: Data Preparation

Create a new spreadsheet and rename it "Data". To combine the Extreme Vertices design with the response data, right-click on the spreadsheet and select Import from Multiple Spreadsheets. In the Multiple Spreadsheet Joiner,

select Horizontal concatenation, with Extreme Vertices as the Left Spreadsheet and Responses as the Right Spreadsheet. Click Execute.

Then, go to Data Transformation → Data Manipulation → Select Column(s). Select only the mixture components A, B, C and the four response variables Drug Release, Drug Loading, Emulsification Time, and Droplet Size. Exclude any additional design-information columns.





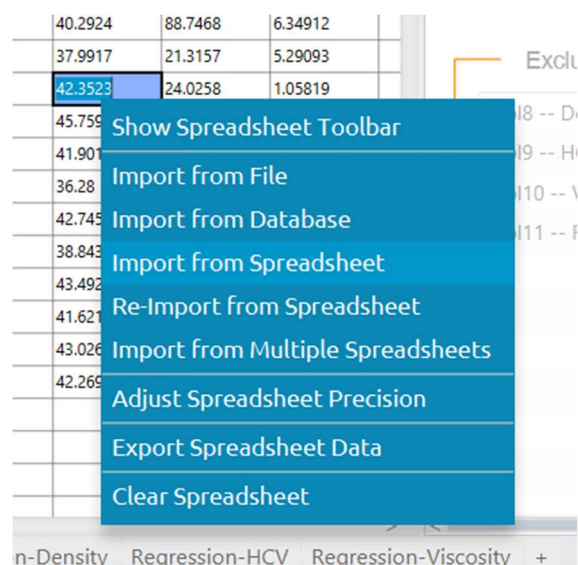
Results :

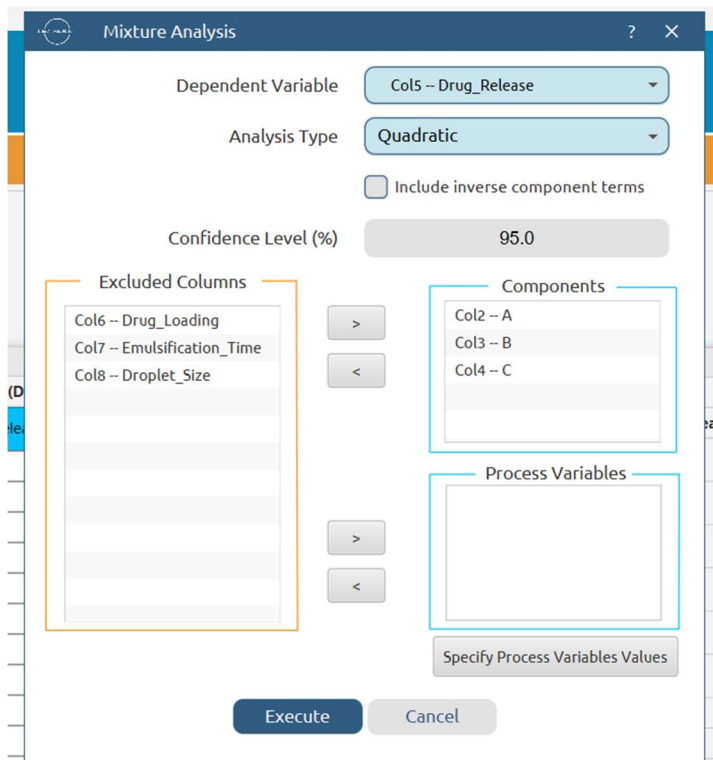
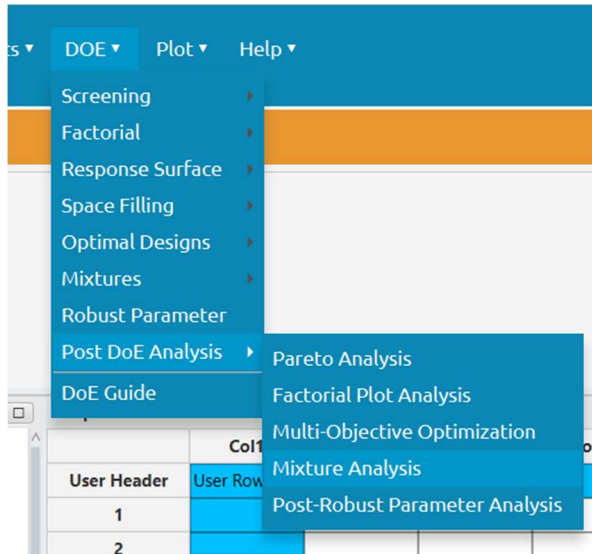
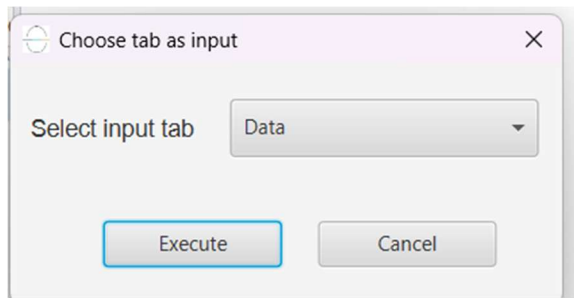
Output								
	Col1	Col2 (D)	Col3 (D)	Col4 (D)	Col5 (D)	Col6 (D)	Col7 (D)	Col8 (D)
User Header	User Row ID	A	B	C	Drug_Release	Drug_Loading	Emulsification_Time	Droplet_Size
1		0.3100000	0.51	0.18	79.91	97.44	45.67	260
2		0.26	0.51	0.23	77.4	97.0	55.67	278.2
3		0.1500000	0.67	0.18	86.14	98.0	41.33	248.53
4		0.12	0.67	0.21	87.11	98.44	42.33	253.67
5		0.12	0.65	0.23	88.8	98.0	36.67	245.1
6		0.19	0.5800000	0.23	74.24	97.11	57.0	290.73
7		0.1350000	0.67	0.195	90.64	99.0	36.0	219.9
8		0.2850000	0.51	0.2050000	85.89	98.33	45.33	257
9		0.2300000	0.5900000	0.18	93.68	100.11	26.33	188.7
10		0.12	0.66	0.22	89.92	100.11	34.33	227.2
11								

Step 4: Drug Release Mixture Analysis

Create a new spreadsheet and rename it "Regression-Drug_Release". Right-click on the spreadsheet, select Import from Spreadsheet, choose the Data tab as input, and click Execute.

Then, go to DOE → Post DOE Analysis → Mixture Analysis. Select Drug Release as the dependent variable and A, B, and C as the mixture components. Select Quadratic as the analysis type, keep the confidence level at 95%, and click Execute to obtain the fitted mixture model and its statistical results.



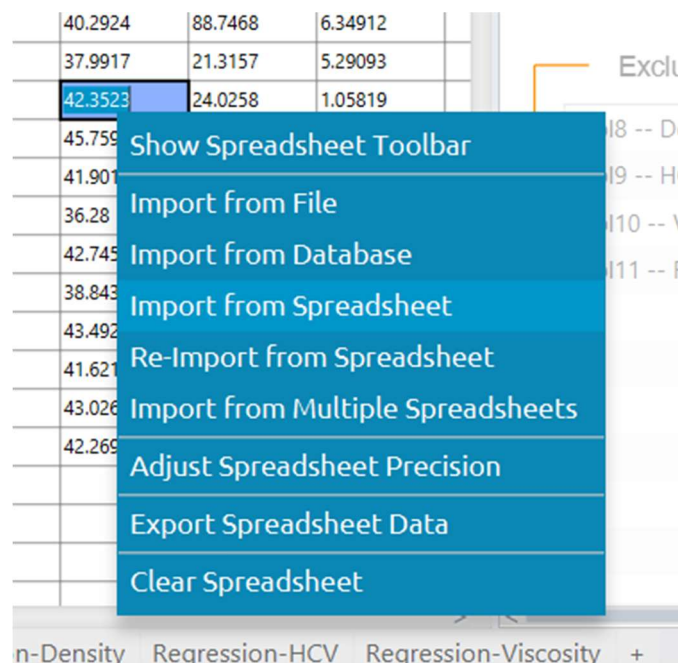


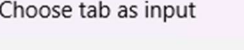
Output										
	Col1	Col2 (S)	Col3 (D)	Col4 (D)	Col5 (D)	Col6 (D)	Col7 (D)	Col8 (D)	Col9 (D)	Col10 (D)
User Header	User Row ID	Variable	Coefficient	Std. Error	Lower CI	Upper CI	Test Statistic	df	p-value	VIF
1		A	-318.5226913	625.3856474	-2054.8716106	1417.8262281	-0.5093220	1.0	0.6373382	4589.6011273
2		B	-324.8531768	431.8963995	-1523.9898209	874.2834673	-0.7521553	1.0	0.4937971	19223.6836663
3		C	-5382.8106049	5358.5266673	-20260.4657404	9494.8445305	-1.0045318	1.0	0.3719597	345501.2771506
4		A*B	466.8600354	892.9499718	-2012.3665428	2946.0866135	0.5228289	1.0	0.6287159	2941.6784925
5		A*C	7726.7985664	8351.9155558	-15461.8364961	30915.4336290	0.9251529	1.0	0.4072483	33777.9323580
6		B*C	8843.3347930	8681.7128644	-15260.9643926	32947.6339786	1.0186164	1.0	0.3659827	332403.3907223
7										

Step 5: Drug Loading Mixture Analysis

Create a new spreadsheet and rename it “Regression-Drug_Loading”. Right-click on the spreadsheet, select Import from Spreadsheet, choose the Data tab as input, and click Execute.

Then, go to DOE → Post DOE Analysis → Mixture Analysis. Select Drug Loading as the dependent variable and A, B, and C as the mixture components. Select Quadratic as the analysis type, keep the confidence level at 95%, and click Execute to obtain the fitted mixture model and its statistical results.





Choose tab as input

Select input tab

Data

Execute Cancel

Mixture Analysis

Dependent Variable: Col6 -- Drug_Loading

Analysis Type: Quadratic

☐ Include inverse component terms

Confidence Level (%): 95.0

Excluded Columns:

- Col5 -- Drug_Release
- Col7 -- Emulsification_Time
- Col8 -- Droplet_Size

Components:

- Col2 -- A
- Col3 -- B
- Col4 -- C

Process Variables:

Specify Process Variables Values

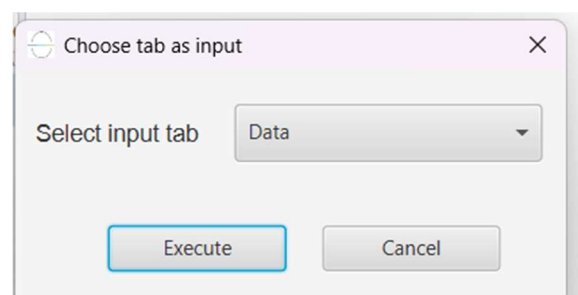
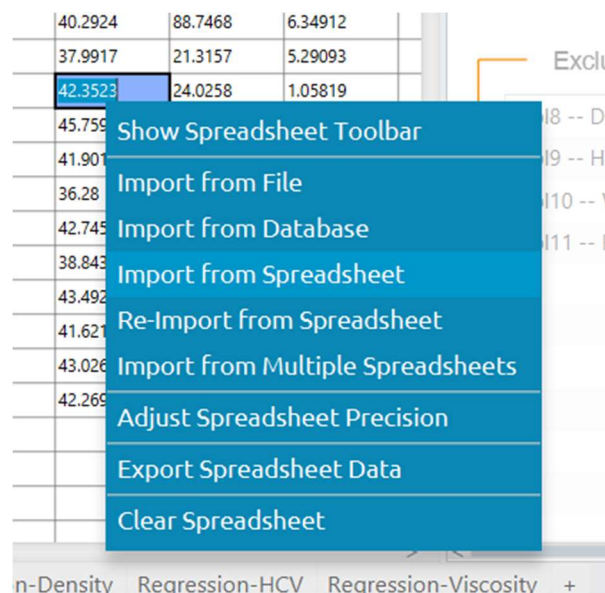
Execute Cancel

[illegible]

Step 6: Emulsification Time Mixture Analysis

Create a new spreadsheet and rename it “Regression-Emulsification_Time”. Right-click on the spreadsheet, select Import from Spreadsheet, choose the Data tab as input, and click Execute.

Then, go to DOE → Post DOE Analysis → Mixture Analysis. Select Emulsification Time as the dependent variable and A, B, and C as the mixture components. Select Quadratic as the analysis type, keep the confidence level at 95%, and click Execute to obtain the fitted mixture model and its statistical results.



Mixture Analysis

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Dependent Variable

Col7 -- Emulsification_Time

Analysis Type

Quadratic

☐ Include inverse component terms

Confidence Level (%)

95.0

Excluded Columns

Col5 -- Drug_Release

Col6 -- Drug_Loading

Col8 -- Droplet_Size

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Components

Col2 -- A

Col3 -- B

Col4 -- C

Process Variables

Specify Process Variables Values

Execute

Cancel

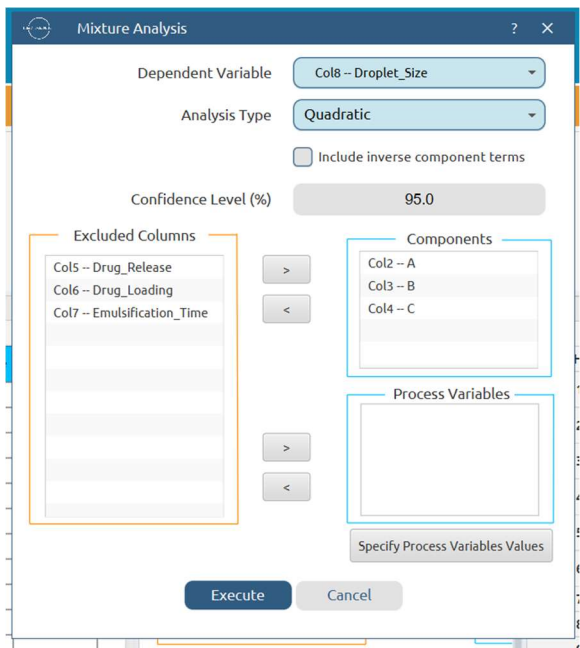
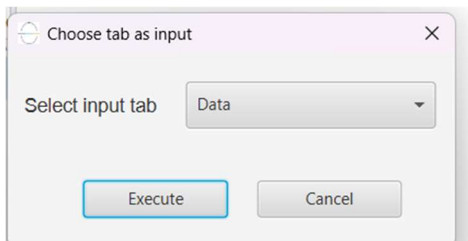
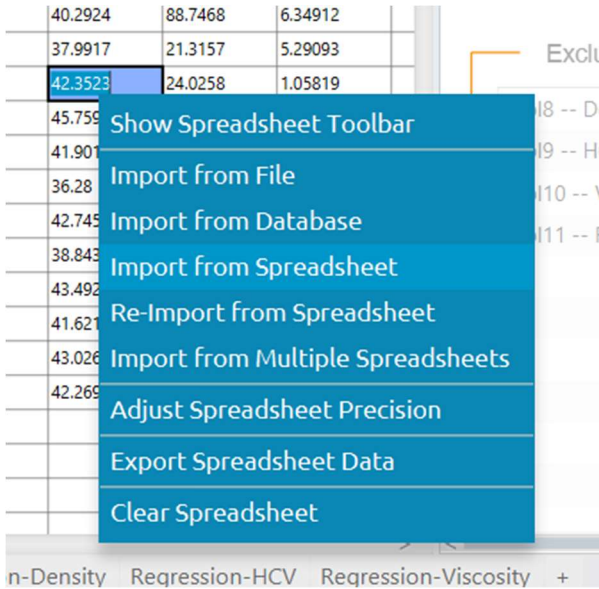
Output											
	Col1	Col2 (S)	Col3 (D)	Col4 (D)	Col5 (D)	Col6 (D)	Col7 (D)	Col8 (D)	Col9 (D)	Col10 (D)	Col
User Header	User Row ID	Variable	Coefficient	Std. Error	Lower CI	Upper CI	Test Statistic	df	p-value	VIF	
1		A	363.0965812	894.7831530	-2121.2197240	2847.4128863	0.4057928	1.0	0.7056639	4589.6011273	
2		B	522.0353299	617.9445014	-1193.6536563	2237.7243161	0.8447932	1.0	0.4457940	19223.6836663	
3		C	5073.7012664	7666.8203163	-16212.8044719	26360.2070046	0.6617739	1.0	0.5442948	345501.2771506	
4		A*B	-944.6838931	1277.6062172	-4491.8874209	2602.5196348	-0.7394171	1.0	0.5006965	2941.6784925	
5		A*C	-5683.0123241	11949.6719601	-38860.6205444	27494.5958962	-0.4755789	1.0	0.6591819	33777.9323580	
6		B*C	-8545.9629733	12421.5361240	-43033.6761417	25941.7501951	-0.6879957	1.0	0.5292759	332403.3907223	
7											

Step 7: Droplet Size Mixture Analysis

Create a new spreadsheet and rename it "Regression-Droplet_Size". Right-click on the spreadsheet, select Import from Spreadsheet, choose the Data tab as input, and click Execute.

Then, go to DOE → Post DOE Analysis → Mixture Analysis. Select Droplet Size as the dependent variable and A, B, and C as the mixture components. Select

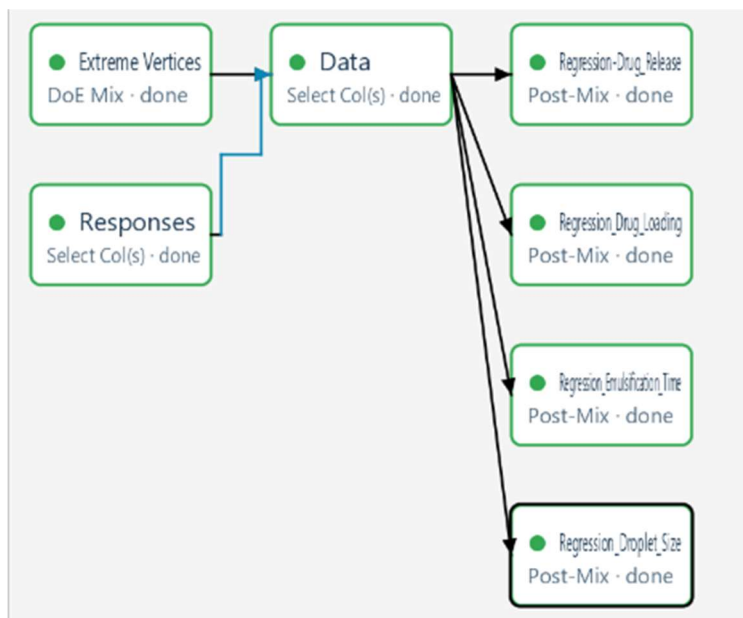
Quadratic as the analysis type, keep the confidence level at 95%, and click Execute to obtain the fitted mixture model and its statistical results.



Output										
	Col1	Col2 (S)	Col3 (D)	Col4 (D)	Col5 (D)	Col6 (D)	Col7 (D)	Col8 (D)	Col9 (D)	Col10 (D)
User Header	User Row ID	Variable	Coefficient	Std. Error	Lower CI	Upper CI	Test Statistic	df	p-value	VIF
1		A	2474.2111569	3019.2643045	-5908.6104421	10857.0327559	0.8194748	1.0	0.4585334	4589.6011273
2		B	1996.4631698	2085.1284126	-3792.7814046	7785.7077442	0.9574773	1.0	0.3925473	19223.6836663
3		C	20202.2143540	25870.1304703	-51624.7827565	92029.2114645	0.7809089	1.0	0.4784882	345501.2771506
4		A*B	-4215.3260209	4311.0231054	-16184.6450194	7753.9929776	-0.9778018	1.0	0.3835373	2941.6784925
5		A*C	-26628.9281469	40321.7448606	-138580.0392910	85322.1829971	-0.6604111	1.0	0.5450834	33777.9323580
6		B*C	-32911.0299360	41913.9547966	-149282.8245629	83460.7646908	-0.7852046	1.0	0.4762327	332403.3907223
7										

Final Isalos Workflow

The final workflow is presented below :



References

Reddy, K. A. N., et al. (2020). *Design and Development of Darunavir Loaded Self Micro Emulsifying Drug Delivery System using Extreme Vertices Mixture Design in a Quality by Design Framework*. Indian Journal of Pharmaceutical Education and Research, 54(2), 337–348.

<https://doi.org/10.5530/ijper.54.2.39>